PRACTICAL BIOCHEMISTRY LECTURE • Laboratory Diagnostics

Qualitative Analysis of Proteins & Amino Acids

Systematic identification schemes, biochemical principles, and colorimetric test procedures.

INSTRUCTOR
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FOCUS AREAS
Amino Acid Identification

DIAGNOSTIC WORKFLOW

Scheme for Protein & Amino Acid Detection

Laboratory Flowchart

1. General Protein Tests

Biuret Test: Detects peptide bonds

Lead Acetate Test: Sulfur group presence

Sulphosalicylic Acid: Protein precipitation

Confirms intact protein macromolecule structure.

2. Specific Amino Acid Assays

Xanthoproteic: Aromatic rings (Tyr, Trp)

Hopkins-Cole: Indole ring (Tryptophan)

Millon's Test: Phenolic group (Tyrosine)

Sulphur Test: Cysteine & Cystine detection

Ninhydrin: Free α-amino groups

Identifies individual amino acid side chains.

3. Individual Protein ID

Acetic Acid Test: Casein precipitation

Heat Coagulation: Albumin & Globulin

Salt Saturation: Half (Globulin) / Full (Albumin)

Separates proteins by solubility & heat sensitivity.

AROMATIC AMINO ACID IDENTIFICATION

Xanthoproteic Test: Reaction Principle

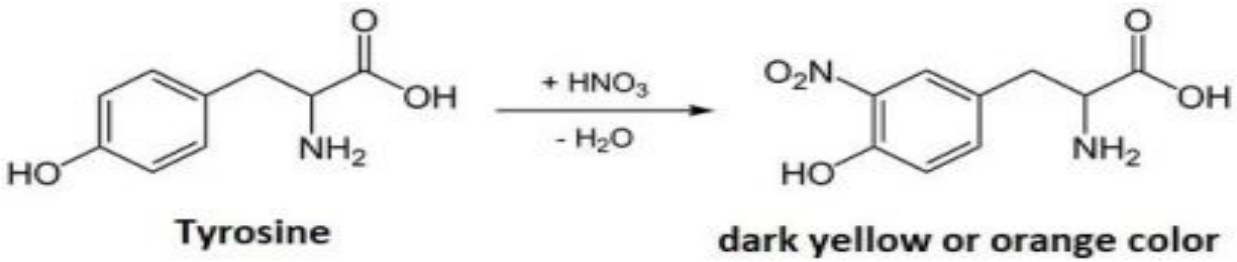
TARGET FUNCTIONAL GROUP

Aromatic Rings (Phenyl & Indole Groups)

Detects Tyrosine and Tryptophan. Phenylalanine requires prolonged heating due to ring stability.

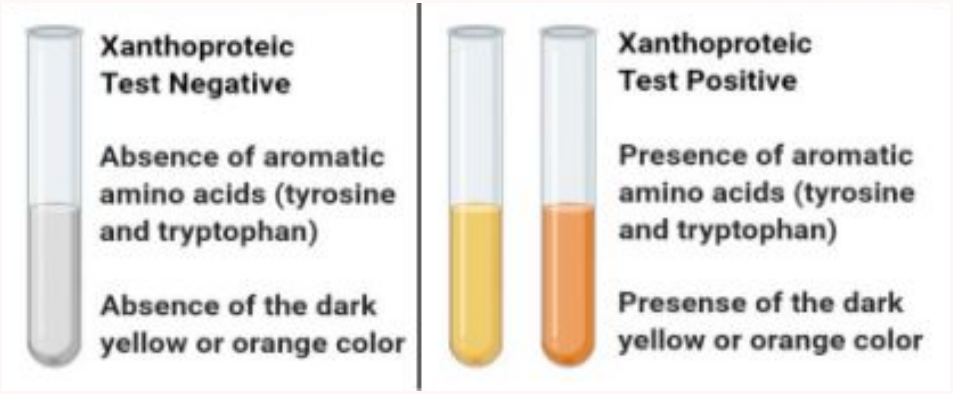
1. **Nitration Step (Conc. HNO_3)**
   Heating yields yellow-colored nitro derivatives of aromatic rings.

2. **Alkalization Step (40% NaOH)**
   Addition of alkali forms tautomeric salts, intensifying color to dark orange.



Tyrosine

dark yellow or orange color



Positive Outcome

Dark Yellow or Orange Color

Confirms presence of activated aromatic amino acid residues in the test sample.

EXPERIMENTAL PROCEDURE

Xanthoproteic Protocol & Limitations

Uses

This is a biochemical test for the detection of proteins and amino acids,allows the differentiation of aromatic amino acids from non-aromatic amino acids

Step 1: Acid Addition

Combine 1 mL sample solution with 1 mL concentrated Nitric Acid (HNO_3).

Step 2: Cooling Phase

Cool to room temperature. Protein denaturation forms a white precipitate.

Step 3: Base Addition

Add 1 mL 40% NaOH. Observe shift to intense yellow or dark orange.

Unresponsive Phenylalanine: Stable benzene ring requires prolonged heating.
Qualitative Only: Cannot quantify protein concentration. Corrosive HNO_3 causes skin staining.

SPECIFIC INDOLE RING DETECTION

Hopkins–Cole Test: Tryptophan Assay

TARGET MOLECULE

Indole Ring of Tryptophan

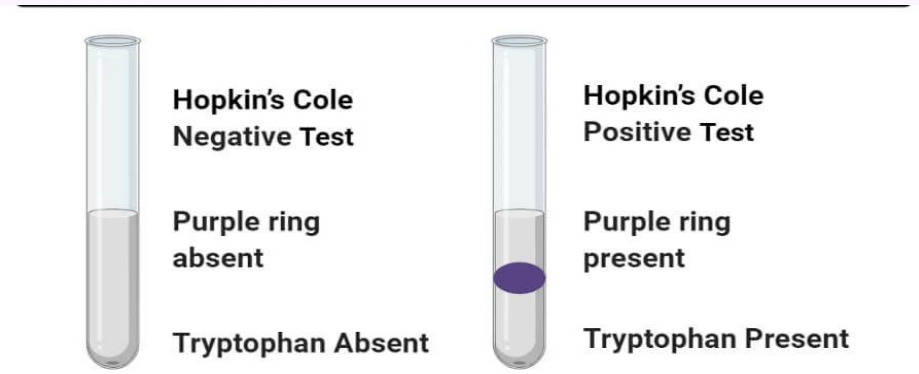
Also known as the Glyoxylic Acid Test. Highly specific for free or bound tryptophan residues.

Chemical Condensation Mechanism

- **Reagent:** Glyoxylic acid (prepared by exposing glacial acetic acid to sunlight).
- **Coupling:** Glyoxylic acid bridges two indole rings via aldehyde oxidation.
- **Dehydration:** Acid-catalyzed loss of water yields a violet pigment.

Reaction

Tryptophan+Glyoxylic acid→Condensation product + H2O→ Violet-coloured compound + H2O



Positive Hallmark

Violet/Purple Ring at Junction

Forms strictly at the interface between the aqueous sample layer and concentrated H_2SO_4.

METHODOLOGY & ANALYTICAL PITFALLS

Hopkins–Cole Protocol & Interferences

Uses of Hopkin's Cole Test

•The test is used for the detection of proteins and amino acids in a sample.
•The test is a simple and easy-to-perform test that helps to identify tryptophan from other
•amino acids

Standard Test Procedure

1. **Sample Mixture**
   Add 2 mL sample solution + 2 mL Hopkin's Cole reagent into a test tube.
2. **Acid Layering**
   Incline tube; slowly add conc. H_2SO_4 down the side to form two distinct layers.
3. **Interface Observation**
   Inspect junction for purple ring. **Do not shake tube**

Agitation disrupts the interface layer and destroys ring visibility.

Chemical Interferences

Inhibited by **nitrites, chlorates, nitrates, and excess chlorides**. These oxidation agents prevent proper aldehyde-indole condensation.

Gelatin Negative Control

Gelatin yields a negative result despite being a pure protein, because its primary sequence naturally lacks tryptophan residues.

PHENOLIC RESIDUE IDENTIFICATION

Millon's Test: Tyrosine Detection

TARGET FUNCTIONAL GROUP

Phenolic (–OH) Group of Tyrosine

Millon's reagent specifically nitrates phenolic hydroxy rings to form colored mercury complexes.

Reagent Formulation

Mercuric (Hg^{2+}) and Mercurous (Hg^{+}) nitrates dissolved in concentrated nitric acid. Highly toxic hazard.

Test Execution

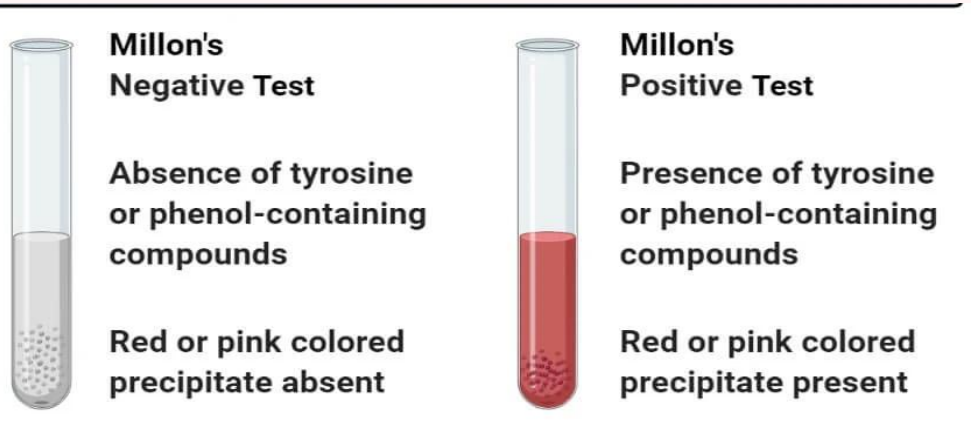
- Mix 2 mL sample with 2 mL Millon's reagent. Heat in a boiling water bath for 2 minutes.

•Uses

- Detects tyrosine-containing proteins.
- Helps differentiate tyrosine from other amino acids.
- Can be used for proteins such as casein

•Limitations

- Not protein-specific: Other phenolic compounds can also give a positive result.
- Cannot detect all proteins: Proteins without tyrosine (e.g., gelatin) may give a negative result.
- Highly toxic: Contains mercuric/mercurous ions and nitric acid, which are toxic and corrosive.
- Possible interference: Heavy metals may cause protein precipitation or denaturation, making results difficult to interpret.



Positive Outcome

Red / Pink Precipitate

Initial white precipitate converts to brick-red complex upon water bath heating.

SULFUR-CONTAINING AMINO ACIDS

Sulphur (Lead Acetate) Test

Alkaline Hydrolysis Principle

Reaction

$Cysteine + 2NaOH \rightarrow Serine + Na_2S + H_2O$

$Pb(CH_3COO)_2 + NaOH \rightarrow Pb(OH)_2 + 2CH_3COONa$

$Pb(OH)_2 + 2NaOH \rightarrow Pb(ONa)_2 + 2H_2O$

$Na_2S + Pb(ONa)_2 + H_2O \rightarrow PbS\ (black\ precipitate) + 4NaOH$

Boiling with 40% NaOH releases inorganic sulfide ions (S^{2-}). Addition of lead acetate precipitates insoluble **Lead Sulfide (PbS)**.

Limitations

- Cannot detect methionine: Methionine does not release sulfide under test conditions, so no black precipitate forms.
- Not quantitative: Only indicates the presence or absence of sulfur-containing amino acids.
- Possible interference: Impurities or sulfides may cause false-positive results.
- Harsh conditions: Strong NaOH and boiling may cause degradation or charring, making results difficult to interpret

Uses

•The test is used to detect sulfur-containing amino acids like cysteine and cystine.
•It helps to distinguish between different groups of amino acids.
Clinical Significance: Cystine detection in urine aids cystinuria kidney stone diagnosis.

Cysteine & Cystine

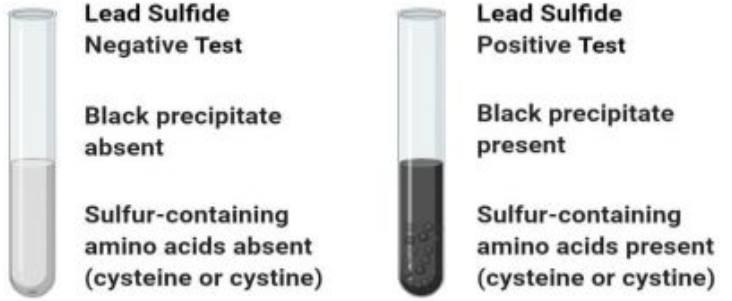
POSITIVE

Contains labile sulfhydryl (-SH) or disulfide (-S-S-) bonds that break readily in alkali.

Methionine

NEGATIVE

Contains a stable thioether bond (-C-S-C-) resistant to NaOH hydrolysis. Yields no black PbS.



UNIVERSAL ALPHA-AMINO ACID DETECTION

Ninhydrin Test: Free Amino Group Assay

REACTION MECHANISM

Oxidative Deamination & Decarboxylation

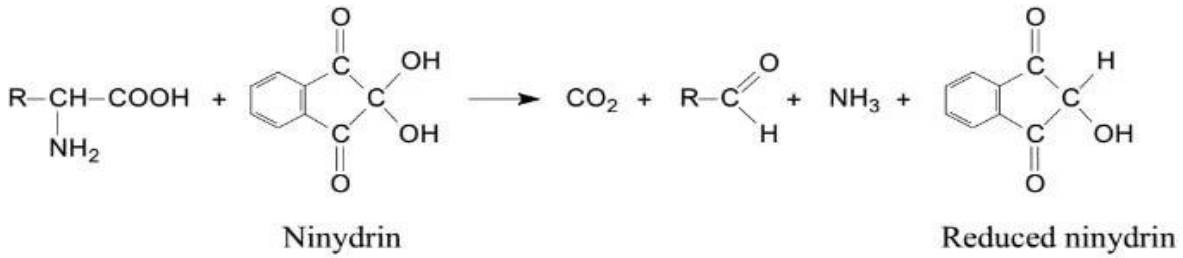
Ninhydrin oxidizes free α-amino groups, releasing NH_3 and CO_2 while forming reduced hydrindantin.

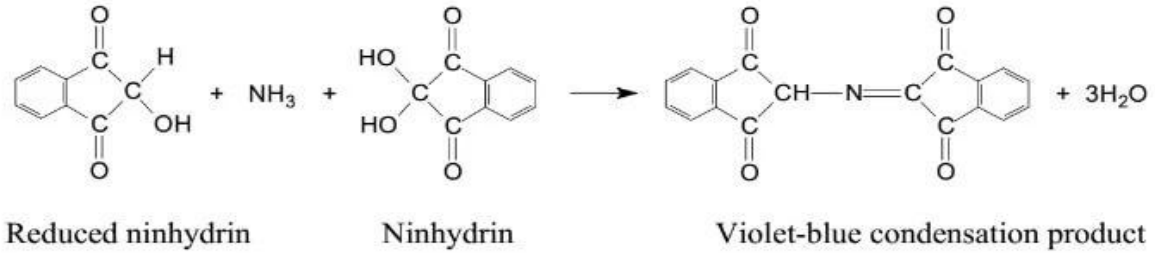
Primary Amine

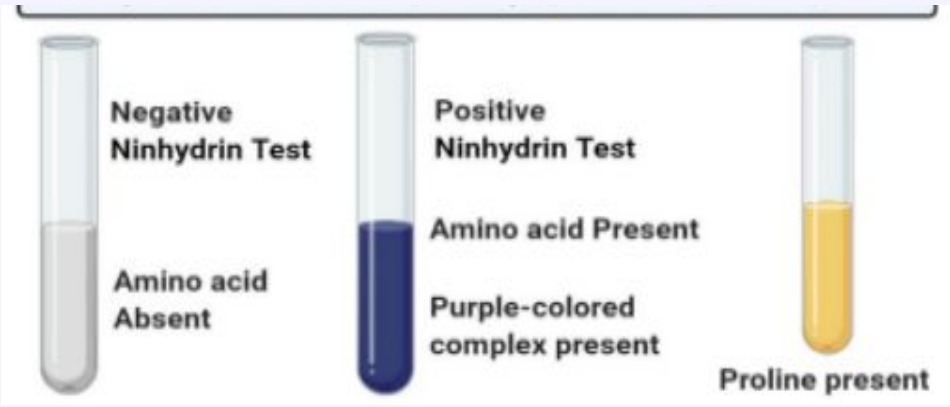
Yields deep blue-purple complex (**Ruhemann's Purple**).

Proline Anomaly

Secondary imino ring forms a unique **yellow-orange** adduct.







Positive Outcome

Ruhemann's Purple Complex

Absorbance at ~570 nm allows quantitative spectrophotometric analysis.

PRACTICAL IMPLEMENTATION

Ninhydrin Applications & Limitations

Forensic Fingerprinting

Reacts with trace amino acids in sweat residue on porous surfaces.

Chromatography Spray

Acts as a chromogenic developer for paper & TLC amino acid spots.

Non-Specific Reaction

Reacts with free amines, urea, and ammonia nitrogen (false positives).

Limitations

- **Low selectivity: Reacts with amino acids, ammonia, urea, and other free amines → may cause false positives.**
- **Proline gives a different color: Produces yellow-orange instead of Ruhemann's purple.**
- **Contamination affects results: Sweat, soil, or other residues may contain amino compounds and interfere.**
- **Safety/practical issues: Some solvents may be flammable, irritating, or cause background staining**
- **Reduced Sensitivity with Large Proteins**

Intact macromolecules suffer steric hindrance, restricting ninhydrin access to internal amino groups.

QUICK REFERENCE TABLE

Summary Matrix of Qualitative Assays

Test Name	Target Group	Key Reagents	Positive Outcome
Xanthoproteic	Aromatic rings (Tyr, Trp)	Conc. HNO_3 + 40% NaOH	**Dark Yellow / Orange**
Hopkins–Cole	Indole ring (Tryptophan)	Glyoxylic acid + Conc. H_2SO_4	**Violet / Purple Interface Ring**
Millon's Test	Phenolic group (Tyrosine)	$Hg(NO_3)_2$ in HNO_3	**Red / Pink Precipitate**
Sulphur Test	Sulfhydryl (Cys, Cystine)	40% NaOH + Lead acetate	**Black Precipitate (PbS)**
Ninhydrin	Free α-amino groups	0.35% Ninhydrin in Ethanol	**Ruhemann's Purple Complex**